



### **Semester 2 Examinations 2016/2017**

**Exam Code(s)** 3BMS2, 3BS9, 4BME1, 4BPT1, 4FM2, 1OA1  
**Exam(s)** 3<sup>rd</sup> Science, 4<sup>th</sup> Science, 4<sup>th</sup> Financial Mathematics and Economics, 4<sup>th</sup> Mathematics and Education,

**Module Code(s)** MP491  
**Module(s)** Nonlinear Systems

Paper No. 1  
Repeat Paper No

External Examiner(s) Professor P. Martin  
Internal Examiner(s) Dr P. Piironen  
Professor M. Tuite

**Instructions:** Marks are awarded for Question 1 in Section A **and** the TWO best questions in Section B.  
All questions carry the same weight.

**Duration** 2 hours  
**No. of Pages** 5  
**Discipline(s)** Applied Mathematics  
**Course Co-ordinator(s)**

**Requirements:**

Release in Exam Venue Yes ☒ No ☐

MCQ Yes ☐ No ☒

Handout None  
Statistical/ Log Tables Yes  
Log Graph Paper None  
Other Materials Calculator

**Section A** - Answer all parts (a)-(d). (**5 Marks**/part)

1. (a) Consider the nonlinear ordinary differential equation

$$\frac{dx}{dt} = x(x^2 - 1).$$

Find all equilibrium points and determine their stability.

- (b) Consider the linear system of ordinary differential equations

$$\begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix} = \begin{pmatrix} 3 & -3 \\ 4 & -3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}.$$

Classify the equilibrium point  $(x, y) = (0, 0)$  and sketch the phase-plane portrait.

- (c) Consider the system of nonlinear ordinary differential equations

$$\begin{aligned} \frac{dx}{dt} &= y^2 - 1, \\ \frac{dy}{dt} &= 1 - x^2. \end{aligned}$$

Find and classify the equilibrium points.

- (d) Consider the map

$$x_{n+1} = 3x_n(1 - x_n)$$

for  $n = 0, 1, 2, \dots$ . Find all fixed points and determine if they are stable or unstable.

**Section B** - Answer any two questions.

2. (a) Consider the one-parameter family of nonlinear ordinary differential equations

$$\frac{dx}{dt} = (x - 2) (a - (x - 1)^2),$$

where  $a$  is a real parameter.

- i. Find all equilibrium points.
- ii. Sketch the bifurcation diagram, and indicate the behaviour of non-equilibrium solutions using appropriate arrows.
- iii. Find all bifurcation points and classify them.

**10 Marks**

- (b) Consider the second order differential equation

$$\ddot{z} - (\mu - z^2)\dot{z} + z = 0 \tag{1}$$

- i. Show that (1) can be written as the system of ordinary differential equations

$$\begin{aligned} \dot{x} &= y, \\ \dot{y} &= -x + (\mu - x^2)y. \end{aligned}$$

- ii. Find the values of  $\mu \in \mathbb{R}$  for which the origin  $(x, y) = (0, 0)$  is a spiral.
- iii. Find the value of  $\mu \in \mathbb{R}$  for which the origin  $(x, y) = (0, 0)$  undergoes a Hopf bifurcation.

**10 Marks**

3. Consider the system of ordinary differential equations

$$\begin{aligned}\frac{dx}{dt} &= y - x, \\ \frac{dy}{dt} &= 1 - xy.\end{aligned}$$

(a) Locate and classify all equilibrium points in the associated phase plane.

**4 Marks**

(b) Identify, and sketch separately, the isoclines corresponding to  $k = -1, 0, 1, \infty$ , where

$$\frac{dy}{dx} = k.$$

**12 Marks**

(c) Use all four isoclines found in (b) to illustrate the phase-plane portrait for this system.

**4 Marks**

4. Consider the one-dimensional map

$$x_{n+1} = \frac{ax_n}{1+x_n},$$

where  $n = 0, 1, 2, \dots$  and  $a$  is a parameter.

- (a) Find all fixed points and investigate for what values of  $a$  the model only has non-negative fixed points.

**5 Marks**

- (b) Calculate the stability of the fixed points found in (a) when varying  $a \in (-\infty, \infty)$ .

**5 Marks**

- (c) Sketch the bifurcation diagram for  $-3 \leq a \leq 3$ .

**5 Marks**

- (d) Use the results in (a)-(c) and determine if  $x$  can undergo any bifurcations when  $x > 0$ . Explain your answer.

**5 Marks**